

A Computational Report

Our framework employs a foundation stereo [11] whose GPU memory usage is highly sensitive to input resolution. We evaluate this trade-off on DTU under different input image downsampling ratios, reporting training time, GPU memory, and reconstruction accuracy. As shown in Tab.5, 50% resolution reduces memory usage to 21 GB with only a 0.07 increase in Chamfer Distance compared to full resolution, substantially outperforming the previous state of the art (1.27) [25]. This enables flexible resolution adjustment based on available GPU memory with minimal quality loss. The total training time is at most 37 minutes, including 20 minutes of warm-up, which is slightly longer than GS-based baselines [8, 30] but significantly faster than NeRF-based methods [9, 20], while achieving better performance.

Downsample	GPU Memory	Training Time	CD↓
50%	21GB	31 mins	1.140
66%	34GB	33 mins	1.117
100%	73GB	37 mins	1.069

Table 5: Computational report across downsampling ratios.

B Different Initialization Strategies.

We conduct an additional experiment to analyze the difference in initialization strategies for the 3DGS [12] and 2DGS [7] modules that affect the final reconstruction quality. We compare three sources of point clouds: (1) SfM [23] point clouds obtained from only three views, (2) point clouds generated by MVSNet [1] under sparse-view settings, and (3) point clouds constructed from our chosen epipolar depth [45]. As shown in Fig.5, the point cloud produced from epipolar depth preserves the finest geometric details among all comparisons, demonstrating its effectiveness as an initialization strategy.

This experiment is conducted under the little-overlap setting, and all configurations except for the initialization method follow the same setup aforementioned. The corresponding mesh reconstruction quality for each initialization method is reported in Tab.6. Initializing with epipolar depth yields the best performance overall. However, the point clouds generated by MVSNet also produce results that surpass the previous state-of-the-art method, SolidGS (1.27), indicating that our method remains robust regardless of the specific initialization strategy. Moreover, as shown in Fig.6, although the epipolar-depth initialization produces the finest geometric details, the MVSNet-based initialization also reconstructs sufficiently reasonable geometry, further demonstrating the robustness of our approach.

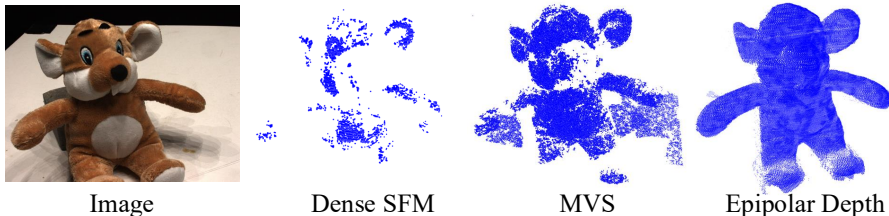


Figure 5: **Generated Initial Pointcloud.** The point cloud generated from epipolar depth is the most dense and preserves the finest geometric details

Init. Strategy	Dense SFM [23]	MVSNet [1]	Epipolar Depth [45]
CD↓	1.341	1.123	1.069

Table 6: **Reconstruction quality of our method under different initialization strategies.**

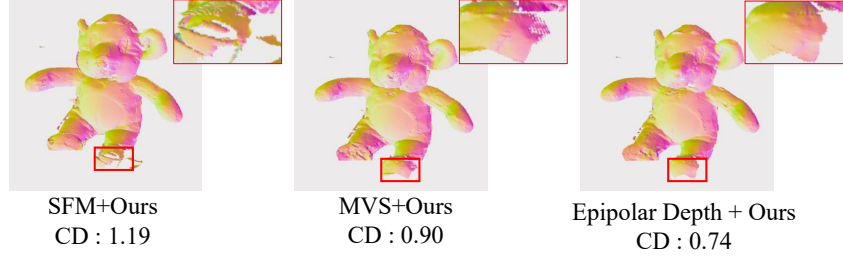


Figure 6: **Visual comparison of the final reconstructed meshes under different initialization strategies.**

C Stereo View Image Rendering.

We compare the quality of stereo view rendering before and after fine-tuning under the stereo view determined in Sec 3.2. Since the stereo views do not have ground truth images, we train the 3DGS under the dense-view setting and use the rendered results as pseudo ground truth for evaluation. We evaluate PSNR and LPIPS, measuring the performance on the occlusion mask M_{occ}^{tr-st} , which is affected by the gradients of our method. As shown in Tab.7, both metrics improve after fine-tuning, demonstrating the effectiveness of our optimization. Moreover, as illustrated in Fig.7, visual artifacts observed in the stereo views before fine-tuning are removed after optimization, resulting in more consistent renderings.

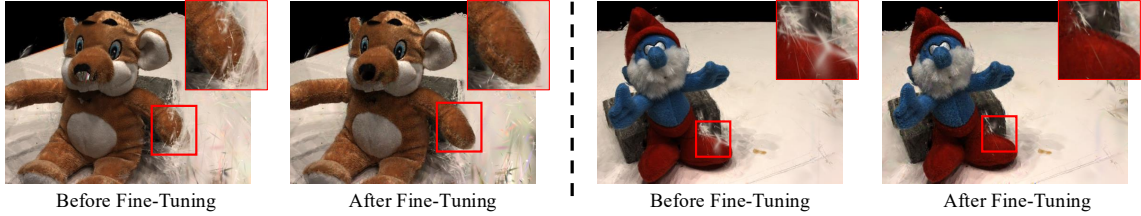


Figure 7: **Effect of fine-tuning on rendering quality.** After the fine-tuning, artifacts are removed, resulting in more accurate stereo renderings.

Fine-Tuning	PSNR↑	LPIPS↓
Before	26.27	0.058
After	26.57	0.054

Table 7: **Rendering quality comparison before and after proposed fine-tuning on stereo view.** The image quality improves after fine-tuning, showing more accurate stereo renderings.